

Difference Between Prim's and Kruskal's Algorithm

Basis	Prim's Algorithm	Kruskal's Algorithm
Approach	Vertex-based	Edge-based
Working	Starts from a vertex	Sorts edges and picks smallest
Structure	Builds single tree	Builds forest then MST
Edge Selection	Visited → Unvisited	Global minimum edge
Cycle Handling	No cycle formed naturally	Cycle checking required
Sorting	Not required	Required
Data Structure	Priority Queue	Disjoint Set (Union-Find)
Efficiency	Better for dense graphs	Better for sparse graphs
Time Complexity	$O(E \log V)$	$O(E \log E)$